

Abdul Rehman Raza

Python Developer | Backend Engineering | Data Science Student

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abdulrehmanraza03.github.io/My-Portfolio/



PROFESSIONAL SUMMARY

Data Science undergraduate (CGPA 3.10/4.00) with demonstrated project engineering experience building asynchronous backend systems, automated data pipelines, and full-stack web applications. Delivered 21+ software solutions spanning SaaS platforms, ML models, and relational database systems. Honhaar Scholarship recipient, elected Student Voice Leader, Google Developer Program member, and NVIDIA Developer Community member.

TECHNICAL SKILLS

Languages: Python • SQL • JavaScript • HTML • CSS

Backend: FastAPI • Flask • SQLAlchemy • REST APIs • Celery • SMTP / Brevo API

Databases: PostgreSQL • SQLite • Redis • RDBMS Schema • SQL Triggers • Stored Procedures

ML & Data: TensorFlow • Scikit-Learn • Pandas • NumPy • DNN • Predictive Classification

Automation: Playwright • Selenium • BeautifulSoup4 • PyYAML

DevOps/Tools: Docker • Docker-Compose • Nginx • Linux CLI • Git • GitHub

FEATURED PROJECTS

SmartScape Pro — Enterprise Web Scraping SaaS [FastAPI • PostgreSQL • Redis • Celery • Playwright • Docker]

- Architected production-ready **multi-tenant SaaS** via **FastAPI** + async **SQLAlchemy 2.0 ORM** with per-tenant **PostgreSQL** data isolation.
- Orchestrated concurrent background scraping via **Celery** + **Redis** message broker; automated cron scheduling via **Celery Beat** with auto-scaling workers.
- Implemented **JWT (RS256)** auth, **Stripe webhooks**, Loguru logging, and full **Docker & Docker-Compose** containerization.

Eco Ride — Smart Fleet & Fuel Analytics Platform [Flask • SQLAlchemy • SQLite • Flask-Login • Flask-Bcrypt]

- Engineered secure multi-user web application with **Flask-Login** + **Flask-Bcrypt** authentication, session management, and role-based access control.
- Built interactive analytics dashboard visualizing GPS route data, fuel consumption trends, and fleet KPIs using modular **SQLAlchemy** backend architecture.

CryptoGuard-AI & AI Diagnostics Platform [TensorFlow • Scikit-Learn • Streamlit • Python]

- Built end-to-end multi-class predictive pipeline via **Scikit-Learn** + **TensorFlow** achieving validated **85% predictive accuracy** on healthcare records with low-latency serialized inference.
- Engineered multi-page **Streamlit** dashboard processing live crypto asset feeds with ML-based risk classification and fraud mitigation.

Charity & Donation Transparency Management System [PostgreSQL • SQL Triggers • Stored Procedures • Views]

- Designed high-performance relational schema with strict multi-entity data validation; formulated server-side **SQL Triggers** and **Stored Procedures** for zero-lag real-time financial tracking.
- Abstracted complex reporting logic into optimized **SQL Views** using advanced multi-table **Joins** — improving query throughput and audit transparency.

EngageBot v2.0 — Automated Engagement Engine [Python • Selenium • PyYAML • Pandas • SMTP]

- Engineered concurrent browser automation via **Selenium WebDriver** with runtime **PyYAML** + **Pandas CSV** template injection; deployed **SMTP** pipeline auto-generating daily analytical reports.

ADDITIONAL PROJECTS

- Student Stress Level Predictor
- Stock Market Forecasting System
- Chess Dataset Cleaning & EDA
- Packet Monitoring & Network Security Tool
- Academic Management Dashboard
- Weather Analytics Dashboard
- FFC Pizza Restaurant Website
- Notes Management App
- Diabetes Risk Assessment Dashboard
- Student Performance Analytics Tool
- Police Station Secure Network Architecture
- Wi-Fi Signal Analysis Tool
- Titan Core Strength Platform
- E-Commerce Data Structure Engine
- Interactive 3D Portfolio (Three.js)
- Replit Collections Tool

PROFESSIONAL EXPERIENCE

Freelance Software & Automation Engineer

Independent Contractor | 2024 – Present

- E-Commerce Engineering:** Architected 'Pakistan Fashion Hub' — full commercial platform with structural relational queries, admin inventory dashboards, and dynamic client routing.
- Production Deployment:** Provisioned and deployed 'FFC Restaurant System' on public version-controlled environments with real-time order tracking interfaces.
- Workflow Automation:** Engineered programmatic data extraction workflows via **Selenium** handling automated communication loops and continuous batch processing pipelines.

EDUCATION & LEADERSHIP

BS Data Science — Superior University, Lahore, Pakistan

2024 – 2028

CGPA: 3.10 / 4.00 | Machine Learning • Database Theory • AI Engineering • Data Analytics • Statistical Modeling

Honhaar Scholarship Recipient

- Elected **Student Voice Leader (SVL)** — selected through competitive faculty evaluation panels, technical interviews, and student-wide voting (2024–2027).

CREDENTIALS & COMMUNITIES

- Google Developer Program** — Active member; leverages cloud sandbox environments for ML deployment evaluation and Data Science projects.
- NVIDIA Developer Community** — Member exploring distributed system acceleration and hardware-optimized deep learning models.
- Languages:** Urdu (Native) • English (Working Proficiency) • Hindi (Fluent)